

Product Requirements Document (PRD): Chetan Shelake - Technical Portfolio Showcase

Document Version: 1.1 (Tailored) **Date:** January 29, 2026 **Author:** Manus AI

1. Introduction

1.1. Goal and Vision

The goal of this project is to create an **authoritative, high-performance digital portfolio** that serves as a technical showcase for Chetan Shelake's expertise as a **Backend Developer specializing in Laravel, complex system architecture, and API integration**. The vision is to use the requested interactive features—specifically multi-layer parallax—to visually deconstruct his complex projects, demonstrating the underlying architecture, data flow, and technical depth rather than just the front-end result. The portfolio must clearly communicate his ability to build robust, scalable, and feature-rich backend systems.

1.2. Target Audience

The primary target audience is highly technical and includes:

- Senior Backend/Full-Stack Recruiters:** Individuals seeking proven expertise in the Laravel ecosystem, system design, and complex API integrations (e.g., Payment, WhatsApp, AI).
- Potential Clients/Hiring Managers:** Organizations looking for a developer capable of architecting and delivering large-scale, custom business solutions (e.g., HR automation, service marketplaces).

1.3. Business Objectives

Objective	Description
Technical Authority	Clearly demonstrate expertise in Laravel, API integration, and database optimization through project presentation.
Project Clarity	Ensure that the architecture and complexity of key projects (QwikHom, Office Management System) are immediately understandable.
Conversion	Achieve a 20% click-through rate to the detailed project case studies or live demos, leading to direct contact.

2. Scope and Release Plan

2.1. In-Scope for Minimum Viable Product (MVP)

The MVP will focus on the technical presentation of Chetan’s four key projects.

Feature Category	MVP Features
Core Interaction	Multi-layer parallax scrolling to represent system architecture layers, 3D elements representing data packets/API calls, scroll-based animation triggers.
Content	Focused sections on QwikHom , Office Management System , MM Precise , and Gurudakshina .
Technical Focus	Dedicated visual space for Tech Stack (Laravel, PHP, MySQL, Flutter, specific APIs) and Architectural Diagrams (simplified).
Performance	Optimized for speed and responsiveness, ensuring the interactive elements do not hinder the professional presentation.

2.2. Out-of-Scope (Future Phases)

- Integrated Blog:** A section for technical articles on Laravel best practices or system design.

- **Interactive Code Snippets:** Live, runnable code examples embedded in the project details.
 - **Multi-Language Support:** Localization for broader reach.
-

3. User Stories and Personas

3.1. Persona: The Senior Backend Recruiter

Name: Sarah Lee **Goal:** To validate Chetan’s claimed expertise in complex backend development and system design.

User Story	Priority
As a recruiter, I want to see a clear, visual breakdown of the tech stack for each project, so that I can quickly match his skills (e.g., AWS Face Recognition API, Laravel 12, Flutter Backend) to the job description.	High
As a recruiter, I want to understand the <i>scale</i> and <i>complexity</i> of the systems he built (e.g., 29-table database, real-time notifications), so that I can assess his capability for senior roles.	High
As a recruiter, I want to easily access his GitHub and LinkedIn profiles, so that I can verify his activity and professional history.	High

3.2. Persona: The Potential Client (Business Owner)

Name: David Chen **Goal:** To find a developer who can solve complex business problems with reliable, custom software.

User Story	Priority
As a client, I want to see how he integrated specific external services (e.g., WhatsApp API, Payment Gateways), so that I can be confident he can handle my custom integration needs.	High
As a client, I want to experience a professional, high-quality website, so that I can trust the quality of the software he delivers.	Medium

4. Detailed Feature Requirements (MVP)

4.1. Core Feature: Parallax Scrolling Architectural Journey (P-1)

The parallax effect must be re-contextualized to represent the layers of a software system.

Requirement ID	Description	Priority	Effort	Impact
REQ-P-1.1	Architectural Parallax: Implement a three-layer parallax: Foreground (Project UI Mockup), Mid-ground (Key Technical Features/Metrics), and Background (Abstract 3D elements representing data flow, API calls, or microservices).	High	Medium	High
REQ-P-1.2	3D Element Context: The 3D elements (geometric shapes, particles) MUST be styled to convey a technical, data-driven aesthetic. They should visually react to the scroll, symbolizing the flow of data packets between the Flutter App, Laravel Backend, and MySQL Database.	High	Medium	High
REQ-P-1.3	Staggered Technical Reveal: As the user scrolls into a project section, the project's technical details (e.g., "29-Table Optimized Database," "Real-time Firebase Integration," "AI-Driven Answer Evaluation") MUST animate in with a smooth, staggered effect, emphasizing the technical complexity.	High	Medium	High
REQ-P-1.4	Tech Stack Icons: Prominent, high-resolution icons for key technologies (Laravel, PHP, MySQL, Flutter, OpenAI, AWS) MUST be displayed for each project.	High	Low	High
REQ-P-1.5	Performance Target: The interactive elements MUST be optimized to maintain a consistent frame rate of 60 FPS on modern desktop hardware, reflecting the developer's commitment to high-performance code.	High	Medium	High

4.2. Content Requirements (Tailored)

Requirement ID	Description	Priority
REQ-CON-2.1	Project Summary: Each project section MUST include a brief, bulleted summary focusing on the backend challenge and the solution implemented (e.g., “Developed complex salary calculation algorithms and multi-guard authentication for HR system”).	High
REQ-CON-2.2	API Integration Highlight: Dedicated text blocks MUST highlight the specific external APIs integrated (e.g., Twilio/Aisensy for WhatsApp, ICCI/PayPhi for Payments, AWS Face Recognition, OpenAI GPT-4.5).	High
REQ-CON-2.3	Contact Information: The contact section MUST prominently feature the GitHub, LinkedIn, and email links.	High

5. Design and Technical Considerations

5.1. Design Aesthetic

The design should be **professional, dark-themed, and technically sophisticated**. The aesthetic should lean towards a “developer dashboard” or “system monitoring” look, using clean lines, monospaced fonts, and a color palette of deep blues, blacks, and electric greens/cyans to highlight interactive elements and data points.

5.2. Technology Stack (High-Level)

The stack remains focused on performance and modern front-end presentation of complex data.

Component	Recommendation	Rationale
Framework	React or Next.js	Best for component-based architecture and performance optimization, allowing for complex state management required by the parallax.
3D Rendering	Three.js or Babylon.js	Necessary for creating the custom, data-flow-themed 3D background elements.
Animation	GSAP (GreenSock Animation Platform) with ScrollTrigger	Essential for the precise, high-performance control over the staggered technical reveals and parallax effects.
Styling	Tailwind CSS or Styled Components	Facilitates rapid, utility-first styling consistent with a modern technical aesthetic.

5.3. Performance Optimization Strategy

Given the 3D and animation load, performance is paramount to avoid the portfolio feeling sluggish, which would contradict the developer's focus on efficiency.

- **Mobile Fallback:** A high-priority requirement is a robust mobile fallback that disables all 3D and complex parallax, presenting a static, fast-loading version of the project summaries.
- **LOD (Level of Detail):** Implement techniques to reduce the complexity of 3D models/particles when the user is not actively scrolling or when the viewport is small.

6. Success Metrics

The following KPIs will be tracked to measure the success of the interactive portfolio:

KPI	Measurement	Target
Average Session Duration	Google Analytics	> 2:30 minutes (Indicates successful engagement with the interactive journey)
Project Click-Through Rate	Google Analytics (Event Tracking on “View Case Study” CTA)	> 20%
Scroll Depth	Scroll Tracking Analytics	95% of users reach the Contact section.
Frame Rate Consistency	Developer Tools Monitoring	Consistent 60 FPS on desktop (Primary success metric for technical execution).
Contact Form Submissions	Backend Logs	5+ per month.